Input Optimization: Technical Worksheet

Grade 7 · Worksheet · About 25 minutes
Work with pen and paper. Use color pencils if you like. You do not need a computer or the internet.

Name: _____

Grade/Section: _____

Date: _____

How to use this worksheet

- Do the parts **in order** from top to bottom.
- Each part has a **time** so you know how long to spend.
- Read every instruction before you write. You can do this on your own.

Part 1 – What you will learn (about 3 minutes)

What you will learn: Input optimization means choosing and configuring input methods (keyboard, shortcuts, speech, forms, defaults) so entering data is faster, clearer, and less error-prone.

Do this:

1. **Name an input method** you use (keyboard, touch, voice, mouse, stylus…).

   _____

2. **Circle** an optimization goal: Fewer errors + less effort · More random tapping

Part 2 – Match method to task (about 5 minutes)

Not every task fits every input. Long essays favor keyboards; quick choices may favor buttons; drawing favors stylus/touch. Optimization starts by matching **method → task**.

> **Real-life scenario:** A survey with 40 open text boxes frustrates users who only needed yes/no choices. Redesigning as options reduces typing.

Discuss with a partner or reflect in writing: When is a multiple-choice input more optimized than free typing?

Do this: Circle the better fit.

Task	Better input (circle one)
Write a 500-word essay	Full keyboard · Only emoji buttons
Choose a grade level	Dropdown/buttons · Paragraph box
Sketch a diagram	Stylus/touch · Number pad only

Part 3 – Reduce repeated entry (about 5 minutes)

Optimization removes repeat work: **templates**, autocomplete, copy-paste with care, saved profiles, and defaults. Enter once, reuse safely.

> **Real-life scenario:** Every lab report, students retype the school header. A template with the header already filled saves minutes each time.

Critical thinking: What risk appears if autocomplete fills the wrong saved address?

Do this:

1. **List two repeated inputs** you type often at school.

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2. **Propose one optimization** (template, shortcut, saved default) for each.

Part 4 – Error-proofing input (about 5 minutes)

Good input design **validates** early: required fields, format hints (YYYY-MM-DD), confirm-before-delete. Catching errors at entry is cheaper than fixing later.

> **Real-life scenario:** A form accepts letters in a phone field. Later the calling system fails. A digit-only field would have blocked the mistake.

Discuss with a partner or reflect in writing: Name one confirmation that is worth the extra click.

Do this: Circle **Yes** or **No**.

Control	Helps optimize quality?
Confirm before delete	Yes · No
Hide all error messages forever	Yes · No
Show example format under a date field	Yes · No

Part 5 – Optimization plan (about 5 minutes)

Do this: Pick one school form or homework flow. Sketch a “before” (painful) and “after” (optimized) input path.

Write one metric you would use to judge success (time, errors, frustration).

Part 6 – Check what you learned (about 2 minutes)

Checklist. Put a ✓ in the box when it is true for you.

☐ I can explain **input optimization**.

☐ I match input **methods to tasks**.

☐ I reduce repeated entry with templates/defaults.

☐ I value early **validation** and confirms.

☐ I can propose a before/after optimization.

When all boxes are checked, you have reviewed the main ideas from this worksheet.

Worksheet on Input Optimization (Technical). Central topic: Input optimization. Main technology area: Human–computer input methods.